

Contribution Journal: L04_Team8_ITAI1371

Module 04 Lab – Exploratory Data Analysis

Team Member Contributions

Joshua Balderas – Lab Notebook (L04_Team8_ITAI1371)

As a member of this team, Joshua Balderas completed the module 04 lab notebook regarding the Exploratory Data Analysis of our Titanic dataset. He imported the dataset and used `.info()` to look at the types of data stored in each column. This method helped him discover null values in the Age and Cabin columns. After using `.describe()` he learned that the average age of passengers on the Titanic was around 29 years old and that only 38% of passengers survived.

He produced multiple charts using Seaborn to visualize our data. When observing the survival distribution countplot, he learned that this is considered to be an imbalanced dataset because there were significantly more deaths than survivors. When looking at survival by passenger class, he learned that 1st class passengers had significantly higher chances of survival than passengers that were in 3rd class. Next, when analyzing survival by gender, he was reminded of how they “allowed women and children to Board the lifeboats first” because there were significantly more females that survived than males. When analyzing age distribution he learned that there was a spike in survival for young children. In the experimentation portion, he generated a countplot based on Embarked and violinplot that compared Fare by survived. What he learned from these charts is that most passengers that embarked from Cherbourg survived and passengers that survived tended to pay a higher fare. Overall, he had fun producing visualizations that taught me how to extract useful information from raw data before creating a machine learning model.

Marisabel Morales – Reflective Journal (L04Journal_R_Team8_ITAI1371)

In her reflective journal, she focused on her personal learning journey through this module's exploration of Exploratory Data Analysis. Initially, EDA felt abstract to her. She

understood it was about "looking at data," but she didn't grasp how critical it was to the machine learning workflow.

As she worked through the lab, she realized that EDA is the foundation upon which all subsequent modeling decisions are built. By visualizing survival rates by gender, class, and age, she could see clear patterns that would directly inform feature selection. One concept that clicked for me was the idea of an imbalanced dataset. When I saw that only 38% of passengers survived, she understood why accuracy alone can be misleading. A model predicting "died" for everyone would be 62% accurate but useless. This deepened my understanding of why precision, recall, and F1-score matter.

she also struggled initially with the FacetGrid for age distribution, but once she understood the two side-by-side histograms, the insight became clear: young children had higher survival rates. Overall, this lab helped her see EDA as an essential investigative process that shapes the entire modeling strategy.

Katherine Rodriguez – Contribution Journal (L04Journal_C_Team8_ITAI1371)

My name is Katherine Rodriguez and I am the author of this contribution journal. In this journal, I will be summarizing what my team and I did together as a team. Then I will reflect on what I learned about machine learning in terms of complexity and what concepts I have a deeper understanding of.

My Deeper Understanding of ML Concepts

When I first saw the machine learning workflow, I was intimidated. While going through this lab and really taking my time to understand EDA, I realized how every step logically leads you to have a better machine learning model. Not only did EDA show me survival rates based on gender, class, and age, but it can help you decide what features you want to include in your model. For example, I now understand that since Cabin had so many missing values, it would not be a good idea to include it as a feature.

I also have a better understanding of supervised vs. unsupervised learning. Supervised learning is when you have what's called an "answer key". An answer key is where you have labeled data that you know the results. The titanic dataset is used for supervised learning because we have the "Survived" column.

Lastly, I understand overfitting on a deeper level. If you create a model that memorizes the data it will be 100% accurate... on that data. When you input new data, your model will most likely be horrible because it didn't learn any real patterns from the training data. EDA can help prevent overfitting by allowing you to see what patterns will generalize.

Workflow Complexity and Team Collaboration

The machine learning workflow is complex but logical. Everything in this workflow feeds into the next and you may find yourself going back to a step after realizing something in a later step. Our group lab reflected that since we each did a different portion of the lab. Joshua showed his knowledge by physically doing the lab. Marisabel showed knowledge by documenting what she learned about herself while doing the lab. I am showing my knowledge by tying everything together and explaining how our group worked together to learn more about machine learning.

Machine learning is a process. A process that asks you questions and allows you to identify patterns. Once you identify these patterns you can make an educated decision on what to do with that data. I came into this module thinking machine learning was just tossing data into a model and calling it a day. Now, I know better.